# Class 10

Dhruvi Sheth 22294

28 Jan 2026

```r
vector_sample_1 <- c(1:10)
vector_sample_1[-1]
```

```
## [1]  2  3  4  5  6  7  8  9 10
```

```r
status <- c("yes","no")
vector_sample_2 <- sample(status, 10, replace = T)
vector_sample_2
```

```
##  [1] "yes" "yes" "no"  "yes" "no"  "yes" "no"  "yes" "no"  "yes"
```

**Dataframe**

```r
dataframe_sample <- data.frame(vector_sample_1, vector_sample_2)
```

```r
head(dataframe_sample)
```

```
##   vector_sample_1 vector_sample_2
## 1               1             yes
## 2               2             yes
## 3               3              no
## 4               4             yes
## 5               5              no
## 6               6             yes
```

```r
str(dataframe_sample)
```

```
## 'data.frame':    10 obs. of  2 variables:
##  $ vector_sample_1: int  1 2 3 4 5 6 7 8 9 10
##  $ vector_sample_2: chr  "yes" "yes" "no" "yes" ...
```

```r
dataframe_sample[,2]
```

```
##  [1] "yes" "yes" "no"  "yes" "no"  "yes" "no"  "yes" "no"  "yes"
```

```r
#skip the second row
dataframe_sample[-2,-2]
```

```
## [1]  1  3  4  5  6  7  8  9 10
```

### matrices

```r
matrix_sample <- matrix(c(1:10),2,5)
# put 1 to 10 in 2 rows and 5 columns
# by_ rows is false by default

matrix_sample
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
```

```r
matrix_sample_1<- matrix(c(1:10),2,2)
```

```
## Warning in matrix(c(1:10), 2, 2): data length differs from size of matrix: [10
## != 2 x 2]
```

```r
matrix_sample_1
```

```
##      [,1] [,2]
## [1,]    1    3
## [2,]    2    4
```

```r
matrix_sample_2 <- matrix(c(1:10),2,6)
```

```
## Warning in matrix(c(1:10), 2, 6): data length [10] is not a sub-multiple or
## multiple of the number of columns [6]
```

```r
matrix_sample_2
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]    1    3    5    7    9    1
## [2,]    2    4    6    8   10    2
```

### Arrays

```r
array_sample <- array (c(1:10), dim = c(2,5,3))
array_sample
```

```
## , , 1
##
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
##
## , , 2
##
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
##
## , , 3
##
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
```

```r
array_sample_1 <- array (c(1:20), dim = c(2,5,3))
array_sample_1
```

```
## , , 1
##
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
##
## , , 2
##
##      [,1] [,2] [,3] [,4] [,5]
## [1,]   11   13   15   17   19
## [2,]   12   14   16   18   20
##
## , , 3
##
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
```

### List

```r
list_sample <- list(vector_sample_1, dataframe_sample, matrix_sample, array_sample)

## in each index, each element in a list
```

```r
list_sample
```

```
## [[1]]
##  [1]  1  2  3  4  5  6  7  8  9 10
##
## [[2]]
```

```
##    vector_sample_1 vector_sample_2
## 1                1             yes
## 2                2             yes
## 3                3              no
## 4                4             yes
## 5                5              no
## 6                6             yes
## 7                7              no
## 8                8             yes
## 9                9              no
## 10              10             yes
##
## [[3]]
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
##
## [[4]]
## , , 1
##
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
##
## , , 2
##
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
##
## , , 3
##
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3    5    7    9
## [2,]    2    4    6    8   10
```